

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

```
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    "header": {},
    "path": {},
    "query": {
      "area": {
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        "required": false,
        "schema": {
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          "type": "string"
        },
        "chamber": {
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          "required": false,
          "schema": {
            "type": "string"
          },
          "date_from": {
            "nullable": true,
            "required": false,
            "schema": {
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              "type": "string"
            },
            "date_to": {
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            "required": false,
            "schema": {
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              "type": "string"
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            "required": false,
            "schema": {
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            "schema": {
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              "type": "boolean"
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            "required": false,
            "schema": {
              "type": "string"
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            "source": {
            "nullable": true,
            "required": true,
            "schema": {
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              "type": "string"
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                    "fields": {
                      "action_code": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
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                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "alarm_type": {
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                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
                          "type": "string"
                        },
                      "mes_status": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
                          "type": "boolean"
                        },
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                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
                      "occurred_at": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "nullable": false,
                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
                      "parameter_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
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                        },
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                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                          "type": "integer"
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                        "nullable": false,
                        "required": true,
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                          "type": "string"
                        },
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                        "required": false,
                        "schema": {
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                          "type": "integer"
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                        "required": true,
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                          "type": "string"
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                        "schema": {
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                        "schema": {
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                          "type": "integer"
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                        },
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                        "required": false,
                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
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                          "min_length": 1,
                          "type": "string"
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                              "required": false,
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                                            },
                                            {
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                              }
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                          }
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                    }
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                }
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        }
      }
    }
  }
}
```

02 GET /trace

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "parameter": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "wafer": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      }
    },
    "responses": {
      "200": {
        "schema": {
          "items": {
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            "fields": {
              "measured_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "recipe_step_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 1.0,
                  "type": "integer"
                },
              "seq_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 0.0,
                  "type": "integer"
                },
              "value": {
                "nullable": false,
                "required": true,
                "schema": {
                  "type": "number"
                },
              "type": "object",
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                      "required": false,
                      "schema": {
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                          "fields": {
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                                      "type": "integer"
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                                    },
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                                  "type": "array",
                                  "shape": "object"
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                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

03 GET /parameters

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
```

```
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```

04 POST /documents/search

RULES []

```
{request: {body: {"additional_properties": false, "fields": {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer"}}, "type": {"type": "object"}, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": -1.0, "type": "number"}}, "section": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "object"}, "array": true, "shape": "array", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

05 GET /relations/chambers/{chamber_id}

RULES [{"node_count equals len(nodes)", "relationship_count equals len(relationships)"]

```
{request: {body: null, "header": {}, "path": {"chamber_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {"include_production_context": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "type": "object"}, "type": "array"}}, "production_context": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"returned_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "truncated": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

06 GET /agent/runs

RULES []

```
{request: {body: null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": false, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id"
```

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  "nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}},
"approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {
  "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {
  "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nu
  llable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "created_
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    : {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"
      }}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED",
        "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object", "type": "array"}}, "fau
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          ype": "string"}}, "fault_color": {"nullable": false, "required": true, "schema": {"type": "null"}}, "fau
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              lse, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "llm_model": {"nullable": false, "
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                        "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": tr
                          ue, "schema": {"items": {"additional_properties": false, "fields": {"n": {"nullable": false, "required":
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                              rue, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema":
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                                      y"}}, "type": "object", "type": "array", "shape": "array", "422": {"schema": {"additional_properties"
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                                          erties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "unio
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                                                {"type": "string"}}, "type": "object", "type": "array"}}, "type": "object", "shape": "object"}},
                                                07 POST /agent/ask
```

```
RULES [{"fault_code and ground_truth_fault_code are forbidden", "metrology.alarm_result is forbidden"}]
{"request": {"body": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": true, "requ
  ired": false, "schema": {"min_length": 1, "type": "string"}}, "question": {"nullable": false, "required":
    true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object", "header":
      {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields":
        {"answer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confide
          nce": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}},
            "evidence": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"
              chunk_id": {"nullable": false, "schema": {"min_length": 1, "type": "string"}}, "doc_id"
                : {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"
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                            le": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false
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                                            : 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1,
                                              "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type":
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                                                                  lable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "METROLOGY": {"additi
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```

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ma": {"min_length": 1, "type": "string"}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}}, "type": "array"}}, "limit": {"nullable": false, "required": true, "schema": {"type": "string"}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object", "404": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}}
```

08 GET /approvals

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item s": {
            "additional_properties": false,
            "fields": {
              "action_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "EQP_HOLD",
                    "MONITORING",
                    "WARNING"
                  ],
                  "type": "string"
                },
              "action_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "agent_run_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "approval_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "approved_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "approved_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "chamber": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "chamber_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "created_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "decided_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "decided_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "decision_comment": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "equipment": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "equipment_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
              "lot": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "lot_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "predicted_fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
              "reason": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "status": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "APPROVED",
                    "PENDING",
                    "REJECTED"
                  ],
                  "type": "string"
                },
              "type": "object",
              "type": "array",
              "shape": "array"
            }
          }
        }
      }
    }
  }
}
```

09 POST /approvals/{approval_id}/decision

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": {
      "additional_properties": false,
      "fields": {
        "decided_by": {
          "nullable": false,
          "required": true,
          "schema": {
            "max_length": 40,
            "min_length": 1,
            "type": "string"
          }
        },
        "decision": {
          "nullable": false,
          "required": true,
          "schema": {
            "enum": ["APPROVED", "REJECTED"],
            "type": "string"
          }
        },
        "decision_comment": {
          "nullable": true,
          "required": false,
          "schema": {
            "max_length": 1000,
            "type": "string"
          }
        },
        "object": {
          "header": {},
          "path": {
            "approval_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            },
            "query": {},
            "responses": {
              "200": {
                "schema": {
                  "additional_properties": false,
                  "fields": {
                    "action_code": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
                        "type": "string"
                      }
                    },
                    "action_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "agent_run_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approval_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approved_at": {
                      "nullable": true,
                      "required": true,
                      "schema": {
                        "format": "date-time",
                        "type": "string"
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

```
ring"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

10 GET /audit-logs

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {"nullable": true, "required": false, "schema": {"type": "string"}}}, "responses": {"200": {"schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array"}, "shape": "array"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

11 GET /dataset/bounds

RULES []

deferred inventory

12 GET /dashboard/summary

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area": {"nullable": true, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "daily_trend": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"date": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}}, "has_r03_consec": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}, "date_range": {"nullable": false, "required": true, "schema": {"items": {"format": "date", "type": "string"}, "type": "array"}}, "equipment_counts": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}
```

```
{}}, "hierarchy": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "pending_approvals": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, 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13 GET /alarms/{source}/{alarm_id}

RULES []

```
{ "request": { "body": null, "header": {}, "path": { "alarm_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "query": {}, "responses": { "200": {"schema": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nul
```

```
label": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}}, "type": "object"}}, "judgement": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "latest_agent_run_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object"}}
```

14 GET /alarms/paged

RULES []

```
{ "request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "judgement": {"nullable": true, "required": false, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}}, "sensor_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"items": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object", "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object"}}
```

15 GET /traces/catalog

RULES []

```
{ "request": {"body": null, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"areas": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "type": "object", "type": "array"}}, "equipments": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambe
```

```
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```

16 POST /traces/search

RULES []

```
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17 GET /relations/equipment/{equipment_id}

RULES []

deferred inventory

18 GET /documents/{document_id}

RULES []

```
{ "request": { "body": null, "header": {}, "path": { "document_id": { "nullable": false, "required": true, "schema": {"type": "string"}}, "query": {}}, "responses": { "200": { "schema": {"additional_properties": false, "fields": {"chunks": { "nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chunk_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```



```
{ "string": "", "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "content": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}}, "type": "object"}, "type": "array"}], { "doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "source_path": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"type": "string"}}, "version": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string", "type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}]]}

19 POST /agent/runs

RULES []

{"request": {"body": {"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "header": {}, "path": {"query": {}}, "responses": {"202": {"schema": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string", "type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}}]}

20 GET /agent/runs/{run_id}

RULES [delivery_policy_lub204lub77dubc740 ACTION-POLICY-V1 lub638lub658lub774luba70 MOCK-NOTIFY-V1lub740lub2b9lubc778lubc18lub824luo0b7bud655lubc778luae30lub85dluc5c6lub774lub774luba54lub77clubcfubc790lub3d9Kafka/MES MockLuc744 luc5f0lub3d9lub55club2e4; luc2b9lubc778 lucOc1ud0dcclub294 nullLuc774lub2e4]

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21 GET /agent/evaluations
RULES ["MOCK-NOTIFY-V1 golden_flow\ub294 protocol\u744a \uba85\u2dc\u558\uace0 live 5 phase\u640 UNKNO
WN/MANUAL_RETRY_NOT_LIVE\ub97c \ubd84\ub9ac\u55c\ub2e4; \uae30\u874 V1 \ubc0f null Empty \u751\ub2f5\u
c740 \ubcf4\u874\u55c\ub2e4", "fault_5class\u640 golden_flow\ub294 \uc11c\ub85c \ub3c5\ub9bd\u801\u7
78 Empty \u0c1\u0d0c\ub2e4", "\uc6d0\ubcf8 artifact \uacbd\ub85c\u00b7hash\u00b7incident label\u740a \u
cf54\u1c\u558\u9c0 \uc54a\ub294\ub2e4"]
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22 POST /agent/runs/{run_id}/retry
RULES []
deferred inventory
23 GET /agent/runs/paged
RULES []
deferred inventory
24 GET /approvals/paged
RULES []
deferred inventory
25 GET /actions
RULES [{"delivery_policy": "lub204lub77dubc740 ACTION-POLICY-V1", "lub638lub658lub774lub70 MOCK-NOTIFY-V1", "lub740
lub2b9lub778lubc18lub824lub00b7ud655lubc778 luba30lub85d lubc5c6lub774 lub774lub54lub77c lubc0f lubc790lub
b3d9 Kafka/MES Mock", "lub744 lubc5f0lub3d9lub55club2e4; lub2b9lub778 lubc0c1ud0dcub294 null", "lub774lub2e4"]}
{"request": {"body": null, "header": {}, "path": {}, "query": {"action_code": {"nullable": true, "require
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CESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MO
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L_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_
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26 GET /actions/{action_id}
RULES [{"delivery_policy \ub204\ub77d\udc740 ACTION-POLICY-V1 \ud638\ud658\udc774\uba70 MOCK-NOTIFY-V1\udc740
\udc2b9\udc778\ubcb18\ub824\ub00b7\ud655\udc778 \uae30\ub85d \uc5c6\udc774 \udc774\uba54\udc77c \ubcb0f \udc790\ub3d9 Kafka/MES Mock\udc744 \uc5f0\ub3d9\ud55c\ub2e4; \uc2b9\udc778 \uc0c1\ud0dc\ub294 null\udc774\ub2e4"]
{"request": {"body": null, "header": {}, "path": {"action_id": {"nullable": false, "required": true, "sch
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EGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND",
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"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"
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27 POST /actions/{action_id}/deliveries/{channel}/retry
RULES []
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28 POST /internal/actions/{action_id}/delivery
RULES {"FAILED requires error_code non-null and permits provider_message_id null", "SENT requires provide
r_message_id non-null and error_code null"}
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, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_id": {"nullable": false, "requ
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hema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}
}}

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RULES []
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RULES []
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31 POST /analytics/validate

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32 GET /analytics/history

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RULES []
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33 GET /analytics/evaluations

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34 GET /audit-logs/paged

RULES []

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35 GET /health

RULES []

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36 GET /health/ready
RULES ["all checks PASS yields status READY and HTTP 200", "any check FAIL yields status NOT_READY and HT
TP 503"]
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AgentEvaluationResponse

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AgentRunAcceptedResponse

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            "schema": {
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AgentRunActionItem

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AgentRunApprovalItem

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      "schema": {
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        "type": "string"
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      "schema": {
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    },
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    "status": {
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AgentRunAskEvidence

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AgentRunCreateRequest

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AgentRunDetailResponse

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[1-9][0-9]*(ud604\u007ac\u00c1\u00b958\u00d558\u00b958)\u00b38\u00c11c \u00ac80\u00c09\u00c124\u00be44 \u00ee8\u00
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```



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}
```

AgentRunItem

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}
```

AlarmAskEvidence

```
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        "schema": {
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          "type": "string"
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          "required": true,
          "schema": {
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            "type": "string"
          },
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            "required": true,
            "schema": {
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            },
            "type": "object"
          }
        }
      }
    }
  }
}
```

AlarmEvidence

```
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      },
      "source_id": {
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        "required": true,
        "schema": {
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          "type": "string"
        },
        "title": {
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          "required": true,
          "schema": {
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            "type": "string"
          },
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            "required": true,
            "schema": {
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          }
        }
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    }
  }
}
```

AlarmItem

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      "alarm_id": {
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        "alarm_type": {
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          "required": true,
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            "type": "string"
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          "area": {
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            "required": true,
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              "type": "string"
            },
            "chamber": {
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              "required": true,
              "schema": {
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                "type": "string"
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              "chamber_id": {
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}
```

```
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AlarmRef

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AlarmSource

```
{ "enum": ["R03", "SUMMARY", "TRACE"], "type": "string" }
```

AnalysisQueryRequest

```
{ "additional_properties": false, "fields": { "question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object" }
```

AnalysisQueryResponse

```
{ "additional_properties": false, "fields": { "columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "cross_check": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["MATCH", "MISMATCH", "SKIPPED"], "type": "string"}}, "summary": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "group_by": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "metric": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"column": {"nullable": true, "required": false, "schema": {"type": "string"}}, "p": {"nullable": true, "required": false, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "number"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["count", "max", "mean", "median", "min", "percentile", "ratio", "std", "sum"], "type": "string"}}, "type": "object"}}, "metric_result": {"nullable": false, "required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"items": {"additional_properties": false, "fields": {"group": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "value": {"nullable": false, "required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"type": "null"}]}}, "type": "object"}, {"type": "array"}, {"type": "null"}]}}, "nl_query_log_id": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {}, "type": "object"}, "type": "array"}}, "visualization": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "required": true, "schema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "required": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}}, "type": "object" }
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ApprovalDecisionRequest

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{ "additional_properties": false, "fields": { "decided_by": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "REJECTED"], "type": "string"}}, "decision_comment": {"nullable": true, "required": false
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```
, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}
ApprovalItem
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "rules": ["PENDING requires all decision fields null", "APPROVED or REJECTED requires decided_by and decided_at", "only EQP_HOLD actions have approvals"], "type": "object"}
AskDocumentEvidenceAlias
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}
AskToolItem
{"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
AuditLogItem
{"additional_properties": false, "fields": {"actor": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}
AuditLogPageResponse
{"additional_properties": true, "fields": {"event_type_counts": {"nullable": false, "required": true, "schema": {"additional_properties": {"type": "integer"}, "fields": {"type": "object"}}, "event_types": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array"}}, "page": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "type": "object"}
AutoToolCallItem
```

```
{
  "additional_properties": false,
  "fields": {
    "n": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "result_summary": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "s": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["ERROR", "SUCCESS", "TIMEOUT"],
        "type": "string"
      }
    },
    "status": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["ERROR", "SUCCESS", "TIMEOUT"],
        "type": "string"
      }
    },
    "tool_name": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

ChamberGraphResponse

```
{
  "additional_properties": false,
  "fields": {
    "context": {
      "nullable": false,
      "required": true,
      "schema": {
        "additional_properties": false,
        "fields": {
          "adjacent_process_step_ids": {
            "nullable": false,
            "required": true,
            "schema": {
              "items": {
                "type": "string"
              },
              "type": "array"
            }
          },
          "area": {
            "nullable": false,
            "required": true,
            "schema": {
              "enum": ["Etch", "Photo"],
              "type": "string"
            }
          },
          "chamber_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          },
          "equipment_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          },
          "model_code": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          },
          "parameter_ids": {
            "nullable": false,
            "required": true,
            "schema": {
              "items": {
                "type": "string"
              },
              "type": "array"
            }
          },
          "process_step_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          },
          "relation_ids": {
            "nullable": false,
            "required": true,
            "schema": {
              "items": {
                "type": "string"
              },
              "type": "array"
            }
          }
        },
        "type": "object"
      }
    },
    "graph_revision": {
      "nullable": false,
      "required": true,
      "schema": {
        "pattern": "[0-9a-f]{64}",
        "type": "string"
      }
    },
    "node_count": {
      "nullable": false,
      "required": true,
      "schema": {
        "minimum": 0,
        "type": "integer"
      }
    },
    "nodes": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "business_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "label": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"],
                "type": "string"
              }
            },
            "name": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "node_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "properties": {
              "nullable": false,
              "required": true,
              "schema": {
                "additional_properties": false,
                "fields": {
                  "from_node_id": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "relation_id": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "pattern": "^REL-[0-9a-f]{20}",
                      "type": "string"
                    }
                  },
                  "to_node_id": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  }
                },
                "type": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                }
              }
            },
            "type": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            }
          },
          "type": "object"
        },
        "rules": [
          "node_count equals len(nodes)",
          "relationship_count equals len(relationships)"
        ],
        "type": "object"
      }
    }
  },
  "type": "object"
}
```

ChamberRelationResponse

```
{
  "additional_properties": false,
  "fields": {
    "graph_revision": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "nodes": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "business_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "display_name": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "id": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "label": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "properties": {
              "nullable": false,
              "required": true,
              "schema": {
                "additional_properties": false,
                "fields": {
                  "from_node_id": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "relation_id": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "to_node_id": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  }
                },
                "type": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                }
              }
            },
            "type": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            }
          },
          "type": "object"
        },
        "rules": [
          "node_count equals len(nodes)",
          "relationship_count equals len(relationships)"
        ],
        "type": "object"
      }
    }
  },
  "type": "object"
}
```

ChartType

```
{
  "enum": ["bar", "histogram", "line", "table"],
  "type": "string"
}
```

ChatToolCallItem

```
{
  "additional_properties": false,
  "fields": {
    "name": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "result": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "result_summary": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "status": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["ERROR", "SUCCESS", "TIMEOUT"],
        "type": "string"
      }
    },
    "tool_name": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

ComparisonMatrix

```
{
  "additional_properties": false,
  "fields": {
    "downstream": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": "NOT_AVAILABLE",
        "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"],
        "type": "string"
      }
    },
    "history": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": "NOT_AVAILABLE",
        "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"],
        "type": "string"
      }
    },
    "metrology": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": "NOT_AVAILABLE",
        "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"],
        "type": "string"
      }
    },
    "sibling": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": "NOT_AVAILABLE",
        "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"],
        "type": "string"
      }
    },
    "upstream": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": "NOT_AVAILABLE",
        "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

CrossCheck

```
{
  "additional_properties": false,
  "fields": {
    "cypher": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "status": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["MATCH", "MISMATCH", "SKIPPED"],
        "type": "string"
      }
    },
    "summary": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

DeliveryCallbackRequest

```
{
  "additional_properties": false,
  "fields": {
    "channel": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "EMAIL",
          "MES MOCK"
        ],
        "type": "string"
      }
    },
    "completed_at": {
      "nullable": false,
      "required": true,
      "schema": {
        "format": "date-time",
        "type": "string"
      },
      "error_code": {
        "nullable": true,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "event_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "provider_message_id": {
        "nullable": true,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "request_hash": {
        "nullable": false,
        "required": true,
        "schema": {
          "max_length": 64,
          "min_length": 64,
          "pattern": "^[0-9a-f]{64}$",
          "type": "string"
        }
      },
      "status": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": [
            "FAILED",
            "SENT"
          ],
          "type": "string"
        }
      },
      "rules": [
        "SENT requires provider_message_id non-null and error_code null",
        "FAILED requires error_code non-null and permits provider_message_id null"
      ],
      "type": "object"
    }
  }
}
```

DeliveryChannel

```
{
  "enum": [
    "EMAIL",
    "MES MOCK"
  ],
  "type": "string"
}
```

DeliveryResult

```
{
  "additional_properties": false,
  "fields": {
    "action_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "channel": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "EMAIL",
          "MES MOCK"
        ],
        "type": "string"
      }
    },
    "completed_at": {
      "nullable": false,
      "required": true,
      "schema": {
        "format": "date-time",
        "type": "string"
      },
      "duplicate": {
        "nullable": false,
        "required": true,
        "schema": {
          "type": "boolean"
        }
      },
      "error_code": {
        "nullable": true,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "provider_message_id": {
        "nullable": true,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "request_hash": {
        "nullable": false,
        "required": true,
        "schema": {
          "max_length": 64,
          "min_length": 64,
          "pattern": "^[0-9a-f]{64}$",
          "type": "string"
        }
      },
      "status": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": [
            "FAILED",
            "SENT"
          ],
          "type": "string"
        }
      },
      "rules": [
        "SENT requires provider_message_id non-null and error_code null",
        "FAILED requires error_code non-null and permits provider_message_id null"
      ],
      "type": "object"
    }
  }
}
```

DeliveryStatus

```
{
  "enum": [
    "BLOCKED",
    "CANCELED",
    "FAILED",
    "SENDING",
    "SENT",
    "UNKNOWN",
    "WAITING"
  ],
  "type": "string"
}
```

DiagnosisBlock

```
{
  "additional_properties": false,
  "fields": {
    "alternative_hypotheses": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "lower_rank_reason": {
              "nullable": false,
              "required": true,
              "schema": {
                "max_length": 1000,
                "min_length": 1,
                "type": "string"
              }
            },
            "summary": {
              "nullable": false,
              "required": true,
              "schema": {
                "max_length": 1000,
                "min_length": 1,
                "type": "string"
              }
            }
          },
          "type": "object"
        },
        "type": "array"
      },
      "cause_summary": {
        "nullable": true,
        "required": false,
        "schema": {
          "max_length": 2000,
          "type": "string"
        }
      },
      "confidence": {
        "nullable": true,
        "required": false,
        "schema": {
          "maximum": 1.0,
          "minimum": 0.0,
          "type": "number"
        }
      },
      "diagnostic_coverage": {
        "nullable": true,
        "required": false,
        "schema": {
          "type": "string"
        }
      },
      "evidence_synthesis": {
        "nullable": true,
        "required": false,
        "schema": {
          "max_length": 2000,
          "type": "string"
        }
      },
      "limitations": {
        "nullable": false,
        "required": true,
        "schema": {
          "items": {
            "type": "string"
          },
          "type": "array"
        }
      },
      "observations": {
        "nullable": false,
        "required": true,
        "schema": {
          "items": {
            "type": "string"
          },
          "type": "array"
        }
      },
      "origin_assessment": {
        "nullable": true,
        "required": false,
        "schema": {
          "additional_properties": false,
          "fields": {
            "basis": {
              "nullable": false,
              "required": true,
              "schema": {
                "items": {
                  "additional_properties": false,
                  "fields": {
                    "id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "max_length": 160,
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "namespace": {
                      "nullable": false,
                      "required": true,
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DocumentAskEvidence

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      "required": true,
      "schema": {
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        "type": "string"
      }
    },
    "section": {
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      "schema": {
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    },
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GraphEvidence

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GraphNode

```
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}
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GraphQueryRequest

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    }
  },
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```

GraphQueryResponse

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          }
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GraphRelationship

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      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "relation_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "pattern": "^REL-[0-9a-f]{20}$",
        "type": "string"
      }
    },
    "to_node_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

GroupedMetricResult

```
{
  "additional_properties": false,
  "fields": {
    "group": {
      "nullable": false,
      "required": true,
      "schema": {
        "additional_properties": true,
        "fields": {},
        "type": "object"
      }
    },
    "value": {
      "nullable": false,
      "required": false,
      "schema": {
        "type": "union",
        "variants": [
          {
            "type": "integer"
          },
          {
            "type": "number"
          },
          {
            "type": "null"
          }
        ]
      }
    }
  },
  "type": "object"
}
```

HTTPValidationError

```
{
  "additional_properties": true,
  "fields": {
    "detail": {
      "nullable": false,
      "required": false,
      "schema": {
        "items": {
          "additional_properties": true,
          "fields": {
            "loc": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "union",
                "variants": [
                  {
                    "type": "string"
                  },
                  {
                    "type": "integer"
                  }
                ]
              }
            },
            "msg": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            },
            "type": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            }
          },
          "type": "object",
          "type": "array"
        }
      }
    }
  },
  "type": "object"
}
```

HealthResponse

```
{
  "additional_properties": false,
  "fields": {
    "status": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["UP"],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

ImpactScopeBlock

```
{
  "additional_properties": false,
  "fields": {
    "check_required": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "kind": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"],
                "type": "string"
              }
            },
            "relation": {
              "nullable": true,
              "required": false,
              "schema": {
                "type": "string"
              }
            },
            "source_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "type": {
              "type": "array"
            },
            "direct": {
              "nullable": false,
              "required": true,
              "schema": {
                "items": {
          "additional_properties": false,
          "fields": {
            "kind": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"],
                "type": "string"
              }
            },
            "relation": {
              "nullable": true,
              "required": false,
              "schema": {
                "type": "string"
              }
            },
            "source_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "type": {
              "type": "object",
              "type": "array"
            },
            "graph_conflict": {
              "nullable": false,
              "required": false,
              "schema": {
                "default": false,
                "type": "boolean"
              }
            },
            "reason_code": {
              "nullable": true,
              "required": false,
              "schema": {
                "pattern": "^[A-Z0-9_]+$",
                "type": "string"
              }
            },
            "status": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": ["AVAILABLE", "EMPTY"],
                "type": "string"
              }
            },
            "summary": {
              "nullable": true,
              "required": false,
              "schema": {
                "max_length": 2000,
                "type": "string"
              }
            }
          }
        }
      }
    }
  },
  "type": "object"
}
```

ImpactScopeItem

```
{
  "additional_properties": false,
  "fields": {
    "kind": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"],
        "type": "string"
      }
    },
    "relation": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "source_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "type": "object"
    }
  },
  "type": "object"
}
```

MetricPlan

```
{
  "additional_properties": false,
  "fields": {
    "column": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "p": {
      "nullable": true,
      "required": false,
      "schema": {
        "maximum": 100.0,
        "minimum": 0.0,
        "type": "number"
      }
    }
  },
  "type": "object"
}
```



```
"type": "number"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["count", "max", "mean", "median", "min", "percentile", "ratio", "std", "sum"], "type": "string"}}, "type": "object"}
MetrologyAskEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}}, "type": "object"}
MetrologyEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["METROLOGY"], "type": "string"}}}, "rules": ["metrology.alarm_result is forbidden"], "type": "object"}
NIQueryHistoryResponse
{"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}}, "type": "object"}
NIQueryLogItem
{"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}
NIQueryOutcome
{"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}
OriginAssessment
{"additional_properties": false, "fields": {"basis": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"max_length": 160, "min_length": 1, "type": "string"}}, "namespace": {"nullable": false, "required": true, "schema": {"enum": ["ALARM", "CHUNK", "LOT_HIST", "PARAMETER", "RELATION"], "type": "string"}}, "type": "object"}, "type": "array"}}, "compared": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"downstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "history": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "metrology": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "si_bling": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "upstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "type": "string"}}, "scope": {"nullable": false, "required": true, "schema": {"enum": ["CURRENT_CHAMBER", "DOWNSTREAM", "EQUIPMENT_COMMON", "UNDETERMINED", "UPSTREAM"], "type": "string"}}, "type": "object"}}, "type": "object"}
OriginBasisRef
{"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"max_length": 160, "min_length": 1, "type": "string"}}, "namespace": {"nullable": false, "required": true, "schema": {"enum": ["ALARM", "CHUNK", "LOT_HIST", "PARAMETER", "RELATION"], "type": "string"}}, "type": "object"}
ParameterFinding
{"additional_properties": false, "fields": {"direction": {"nullable": false, "required": true, "schema": {"enum": ["ABOVE", "BELOW", "BOTH"], "type": "string"}}, "excursion_ratio": {"nullable": false, "required": true, "schema": {"type": "number"}}, "lot_hist_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"max_length": 20, "min_length": 1, "type": "string"}}, "step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "wafer_scope": {"nullable": false, "required": true, "schema": {"enum": ["ALL", "PARTIAL", "SINGLE"], "type": "string"}}, "type": "object"}
```

ParameterItem

```
{
  "additional_properties": false,
  "fields": {
    "LCL": {
      "nullable": true,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "LSL": {
      "nullable": true,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "TARGET": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "UCL": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "USL": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "area": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "Etch",
          "Photo"
        ],
        "type": "string"
      }
    },
    "ctrl_lower": {
      "nullable": true,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "ctrl_upper": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "name": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "parameter_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "parameter_name": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "spec_lower": {
      "nullable": true,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "spec_upper": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "target_value": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "unit": {
      "nullable": true,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "upper_only": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "boolean"
      }
    },
    "type": "object"
  }
}
```

PostActionObservationBlock

```
{
  "additional_properties": false,
  "fields": {
    "message": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": "\ucd5c\u00885 \uc815\u00801 \ub370\u00774\u00d130\u00uc14b\u00uc5d0\u00b294 \uc870\u00ce58 \uc774\u00ud6c4 \u00uac f5\u00uc815 \u00ad00\u00uce21\u00uc12\u00uc774 \uc5c6\u00uc5b4 \ud6a8\u00uacfc\u00ub97c \ud3c9\u00uac00\u00ud560 \uc218 \uc5c6\u00uc74c",
        "type": "string"
      }
    },
    "status": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": "NOT_AVAILABLE_STATIC_DATASET",
        "type": "string"
      }
    },
    "type": "object"
  }
}
```

PublicAgentRunItem

```
{
  "additional_properties": false,
  "fields": {
    "action_id": {
      "nullable": true,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "agent_run_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "alarm_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "alarm_source": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "R03",
          "SUMMARY",
          "TRACE"
        ],
        "type": "string"
      }
    },
    "approval_id": {
      "nullable": true,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "chamber": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "chamber_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "confidence": {
      "nullable": true,
      "required": true,
      "schema": {
        "maximum": 1.0,
        "minimum": 0.0,
        "type": "number"
      }
    },
    "created_at": {
      "nullable": false,
      "required": true,
      "schema": {
        "format": "date-time",
        "type": "string"
      }
    },
    "deliveries": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "channel": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": [
                  "EMAIL",
                  "MES"
                ],
                "type": "string"
              }
            },
            "status": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": [
                  "BLOCKED",
                  "CANCELED",
                  "FAILED",
                  "SENDING",
                  "SENT",
                  "UNKNOWN",
                  "WAITING"
                ],
                "type": "string"
              }
            },
            "type": "object"
          },
          "type": "array"
        },
        "fault_code": {
          "nullable": true,
          "required": true,
          "schema": {
            "enum": [
              "FOC",
              "MFD",
              "OTH",
              "RFM",
              "TMD"
            ],
            "type": "string"
          }
        },
        "fault_color": {
          "nullable": false,
          "required": true,
          "schema": {
            "type": "string"
          }
        },
        "fault_name": {
          "nullable": false,
          "required": true,
          "schema": {
            "type": "string"
          }
        },
        "latency_ms": {
          "nullable": false,
          "required": true,
          "schema": {
            "minimum": 0.0,
            "type": "integer"
          }
        },
        "llm_model": {
          "nullable": false,
          "required": true,
          "schema": {
            "min_length": 1,
            "type": "string"
          }
        },
        "predicted_fault_code": {
          "nullable": true,
          "required": true,
          "schema": {
            "enum": [
              "FOC",
              "MFD",
              "OTH",
              "RFM",
              "TMD"
            ],
            "type": "string"
          }
        },
        "recommended_action": {
          "nullable": true,
          "required": true,
          "schema": {
            "enum": [
              "EQP_HOLD",
              "MONITORING",
              "WARNING"
            ],
            "type": "string"
          }
        },
        "status": {
          "nullable": false,
          "required": true,
          "schema": {
            "enum": [
              "COMPLETED",
              "FAILED",
              "RUNNING",
              "WAITING_APPROVAL"
            ],
            "type": "string"
          }
        },
        "tools": {
          "nullable": false,
          "required": true,
          "schema": {
            "items": {
              "additional_properties": false,
              "fields": {
                "name": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "result_summary": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "s": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "enum": [
                      "ERROR",
                      "SUCCESS",
                      "TIMEOUT"
                    ],
                    "type": "string"
                  }
                },
                "status": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "enum": [
                      "ERROR",
                      "SUCCESS",
                      "TIMEOUT"
                    ],
                    "type": "string"
                  }
                },
                "tool_name": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "type": "object"
              },
              "type": "array"
            },
            "type": "object"
          }
        }
      }
    }
  }
}
```

PublicApprovalDecision

```
{
  "enum": [
    "APPROVED",
    "REJECTED"
  ],
  "type": "string"
}
```

PublicApprovalItem

```
{
  "additional_properties": false,
  "fields": {
    "action_code": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "EQP_HOLD",
          "MONITORING",
          "WARNING"
        ],
        "type": "string"
      }
    },
    "action_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "agent_run_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "approval_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "approved_at": {
      "nullable": true,
      "required": true,
      "schema": {
        "format": "date-time",
        "type": "string"
      }
    },
    "approved_by": {
      "nullable": true,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "chamber": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "chamber_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "created_at": {
      "nullable": false,
      "required": true,
      "schema": {
        "format": "date-time",
        "type": "string"
      }
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HYPOTHESIS_GENERATED / AGENT_RUN / C
APPROVAL_REQUESTED / APPROVAL / C
APPROVAL_DECIDED / APPROVAL / C
ACTION_SENT / ACTION / C delivery service
ACTION_SEND_FAILED / ACTION / C delivery service
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AGENT_RUN_FAILED / AGENT_RUN / C